Identify the Most Reliable Players Under Pressure

Round 1 methodology and evidence report | SupeRR Selector Season 2

Decision in one line. I define pressure from the live chase equation, wickets in hand, and late-innings phase, then rank players by how much better they perform in those deliveries than in their own ordinary deliveries, with a sample-size adjustment.

Data and scope

The supplied files contain 883 match records. The model uses 867 matches with standard-innings deliveries: 200,961 deliveries total. Dates span 22 January 2022 to 1 November 2026. The competition file labels the formats T20 and T20I; the analysis uses both formats together because the challenge asks for comparisons across the supplied competitions. It does not assume that every T20 competition is the IPL.

Pressure tags 76,933 delivery records (38.3%). One pressure tag is assigned to a delivery when the situation at that point meets the rules below. The first-innings total normally sets the second-innings target; a supplied revised target overrides it. The state is measured immediately before each delivery, so that delivery's runs or wicket cannot make itself a pressure delivery. Overs use legal-ball counts, so wides and no-balls do not consume a ball. Revised legal-ball allocations and revised chase targets are used when present. Super-over innings are excluded.

How pressure is defined

Second innings: pressure applies when the required rate is at least 9.0 runs per over, or after 10 overs when five or fewer wickets remain, or during the final five overs. **First innings:** pressure applies in the final five overs when five or fewer wickets remain. These rules capture a demanding chase, a low-wicket chase, or a constrained finish without using the match outcome to label a situation.

This is a transparent heuristic, not a claim that pressure has one universal threshold. The RRR trigger flags an aggressive equation; wicket scarcity and late overs catch high-stakes situations where the RRR alone may not.

Scoring and eligibility

Batters: strike rate is batter runs per ball faced x 100; dismissal rate is dismissals of the striker (using dismissed-player ID) per ball faced x 100. For each player, ordinary rates use deliveries not tagged pressure. The raw advantage is (pressure SR - ordinary SR) - 0.60 x (pressure dismissal rate - ordinary dismissal rate), in points per 100 balls. Wides are excluded from balls faced; no-balls count as faced.

Bowlers: economy is bowler-charged runs per legal ball x 6; wicket rate is bowler-credited wickets per legal ball x 100. The raw advantage is (ordinary economy - pressure economy) x 100/6 + 0.60 x (pressure wicket rate - ordinary wicket rate), in weighted points per 100 balls. Run outs and retirements are excluded from bowler wickets. Bowler-charged runs exclude byes, leg-byes, and penalty runs; wides and no-ball penalties remain charged.

To avoid one-off spikes, batters need at least 200 ordinary balls, 60 pressure balls, and eight distinct pressure matches. Bowlers need at least 180 ordinary legal balls, 60 pressure legal balls, and eight distinct pressure matches. The raw advantage is reliability-adjusted as **score = raw x n / (n + 60)**, where n is pressure balls. This shrinks thin pressure samples toward zero. Eligible players are ordered by this adjusted score; each list contains unique players.

Rankings to enter on Unstop

Rank numbers below follow the adjusted score. Rates are shown to help audit the ordering; the accompanying CSVs preserve the underlying pressure counts, match counts, adjusted score, and player ID.

Top 10 batters

Rank	Player	Pressure SR	Ordinary SR	Delta SR	Pressure balls	Matches under pressure
1	Mohammad Haris	189.7	116.1	+73.6	97	10
2	Dinesh Karthik	187.2	135.9	+51.3	391	40
3	Fakhar Zaman	168.7	116.8	+52.0	243	16
4	Mohammad Nabi	161.3	108.6	+52.7	269	29
5	Najibullah Zadran	181.2	117.6	+63.6	101	11
6	Reeza Hendricks	168.8	127.8	+41.0	382	16
7	Sikandar Raza	157.4	120.6	+36.8	406	30
8	Liam Livingstone	175.9	138.7	+37.3	407	38
9	Jaker Ali Anik	151.9	112.1	+39.8	231	24
10	Pathum Nissanka	158.6	126.2	+32.4	464	24

SR = runs per 100 balls faced. "Ordinary" means the same player's non-pressure deliveries.

Top 10 bowlers

Rank	Player	Pressure Econ	Ordinary Econ	Econ saved	Pressure balls	Matches under pressure
1	Mohammad Wasim	6.11	9.24	+3.13	109	10
2	Akash Madhwal	7.45	10.30	+2.85	124	9
3	Mayank Yadav	7.87	10.03	+2.15	266	17
4	Jamie Overton	7.21	8.90	+1.69	188	18
5	Mohit Sharma	8.73	10.14	+1.41	354	26
6	Gudakesh Motie	7.01	8.54	+1.53	226	18
7	Nitish Reddy	9.36	10.80	+1.43	298	27
8	Corbin Bosch	7.53	9.17	+1.63	188	14
9	Kyle Jamieson	8.16	9.82	+1.66	172	16
10	Kwena Maphaka	8.97	10.46	+1.50	97	8

Economy is runs per over; "Econ saved" is ordinary economy minus pressure economy. A positive number means the bowler conceded fewer runs per over under the model's pressure tags.

Reliability checks and limitations

I required exposure in both ordinary and pressure conditions and at least eight distinct pressure matches, and applied a continuous sample shrinkage rather than treating all raw rates as equally reliable. The comparisons are within-player, which controls for stable individual differences but does not fully adjust for opponents, venues, batting positions, match format, or changing roles. Bowlers and batters may face systematically different pressure contexts.

The supplied delivery fields do not include a validated win-probability or leverage measure. RRR is undefined when no balls remain; those rows are handled without using infinite rates, while the final-five-over / wickets-left rules still apply where relevant. Super-over innings are excluded. The dataset's latest date and completeness are taken from the supplied files, not independently verified.

Rankings are sensitive to the chosen pressure cutoffs and the 0.60 wicket-risk weight. Treat close scores as a cluster, not precise measures of talent. A useful robustness extension would rerun at RRR cutoffs of 8 and 10, plus format-specific baselines, and report rank movement; this report's primary order remains the requested reproducible answer.

Reproduction

Run `python pressure_model.py --data-pack path/to/6af458274c1e8_data_pack.zip --output-dir outputs` with Python 3.10+ and pandas/numpy installed. The model reads the ball-by-ball, matches, and competitions CSVs directly from the supplied archive and writes full leaderboards plus both top-10 CSVs. The companion CSV files include the exact score, component rates, sample sizes, match counts, and player IDs used for the lists. The uploaded report and dropdown selections should use these same rank orders.